

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think> The assistant is thinking about the fact that B cells play a variety of roles in the immune response , including the production of antibodies and the activation of other immune cells . The cluster of differentiation 21 (CD 21) is a protein that is found on the surface of B cells , and it is involved in the process of B cell activation . The genetic modification of B cells to prevent the expression of CD 21 would likely affect the ability of B cells to be activated , which could have a variety of effects on the immune response . The assistant is also thinking about the fact that certain organisms , such as the Epstein -Barr virus (EBV), are able to infect B cells and use them to replicate . The ability of these organisms to infect B cells may be affected by the genetic modification of B cells , which could have a variety of effects on the pathogenesis of the organism . </think><answer> A . Epstein -Barr virus (EBV) </answer>

